

ABNORMALITY REPORT — RCA & CAPA/PAA

Equipment Related Risk: Feed/Effluent Heat Exchanger HE-3301

Problem Statement

Feed/Effluent Heat Exchanger HE-3301 suffered accelerated tube-side fouling reducing duty and raising dP, forcing an unplanned isolation, cleaning and partial rate cut on 21-May-2026.

General Identification & Details

Tag / Plant: HE-3301 | ZCU (Zeta Cracker Unit)

Pre-Risk Level: III (Score 400)

AR Type: OPEDR (Operation) — Uptime

Severity: Moderate

Discipline: STA (Static)

Downtime: 12.0 hours

Date Occurrence: 21 May 2026

Financial Loss: \$183.6k

Date Reported: 22 May 2026

PIC (RCA): STA-02

Immediate Action & Procedure

Immediate Action Taken: Isolate HE-3301, bypass via spare train, hydro-jet clean tube bundle.

Procedure Reference: Procedure No. RCA-P-0050-03 • Form RCA-F-0075-02

DOWNTIME IMPACT

12.0 h

ESTIMATED LOSS

\$183.6k

Feed/Effluent Heat Exchanger HE-3301 — High Fouling — Duty Loss & High dP
Tag / Eq. Class: HE-3301 | Class B

Chronology of Events

Time / Date	Event Description
18-May-2026	Heat-recovery section running; HE-3301 tube-side dP rising (0.4 → 0.9 bar) and cold-outlet temperature falling.
21-May-2026 09:00	Preheat duty dropped ~35% below design; unit rate reduced to protect downstream. Exchanger isolated.
21-May-2026 11:30	Tube bundle inspected — heavy coke/polymer deposition on tube side.
21-May-2026 21:00	Bundle hydro-jet cleaned and returned to service; duty and dP restored, rate ramped back to 30 T/H.

Past Performance Analysis

Historical Data & Evidence: HE-3301 tube-side dP baseline is 0.3–0.4 bar; heat duty design 100%. Over the run, dP climbed steadily while duty fell — a classic fouling curve. No dP-based cleaning trigger existed, so fouling was allowed to progress to a forced cut.

Impact Summary (4W):

- **What:** HE-3301 duty loss & high dP from tube-side fouling
- **When:** 21-May-2026 (09:00)
- **Scope:** Partial rate cut of ZCU heat-recovery section
- **Uptime Loss:** 12.0 hours → 216.0 ton → \$183.6k

Target Setting

ACTUAL CONDITION (Before Improvement) Rate cut due to HE-3301 fouling (duty loss + high dP).	TARGET CONDITION (Project Y) No unplanned rate cut due to HE-3301 fouling (dP < 0.6 bar, duty > 90%).
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Recurrence of this equipment failure mode:
Before = 1 case → Target = 0 case

Root Cause Analysis: Parameter Verification (4P)

Parameter	Result	Evidence / Finding
P1: Tube-side dP above limit	NG	dP reached 0.9 bar vs 0.6 bar limit — heavy fouling.
P2: Heat duty below design	NG	Duty fell to ~65% of design — loss of preheat.
P3: Feed heavy-ends above spec	NG	Heavy-ends 2.4% vs < 1.5% design — increased fouling precursor.
P4: Tube velocity below design	G	Tube velocity 1.9 m/s vs 1.5 m/s min — adequate.
P5: Cooling/heating medium temperature off	G	Shell-side inlet temperature within normal band.

NG = current condition not meeting standard (problem) • G = meets standard

Root Cause Analysis: 4M + 1E Verification

Factor	Result	Evidence / Finding
X1: No dP-based cleaning trigger	NG	Cleaning was time-based only; no dP alarm to trigger action.
X2: Feed filtration performance not trended	NG	Upstream filter dP not monitored, allowing heavy-ends carry-over.
X3: Fouling KPI not on dashboard	NG	No fouling factor / duty KPI tracked between TAs.
X4: Bundle metallurgy inadequate	G	Bundle metallurgy suitable for service.

ROOT CAUSE

Accelerated tube-side coke/polymer fouling driven by higher-than-design feed heavy-ends, allowed to progress by the absence of a dP-based cleaning trigger and fouling KPI, causing duty loss, high dP and a forced rate cut.

Corrective & Pro-Active Action (CAPAA)

Corrective / Pro-Active Action	Plan Date	PIC	Status
X1: Add dP-based cleaning trigger & alarm for HE-3301	22-Jun-2026	STA-02	In Progress
P3: Tighten feed heavy-ends control & filtration	30-Jun-2026	ROT-01	Open
Pro-Active: X1: Apply dP-trigger logic to parallel exchangers HE-3302/3303	22-Jul-2026	STA-02	Open
Pro-Active: P3: Review upstream fractionation cut-point control	31-Jul-2026	ROT-01	Open

Preventive & Risk Analysis

Item / Root Cause	Preventive Action / Countermeasure	Plan Date	PIC
X3: No fouling KPI	Add fouling factor & duty KPI to reliability dashboard	22-Jun-2026	STA-02
X2: Filter not trended	Add upstream filter dP trend & replacement trigger	30-Jun-2026	ROT-01
Add dP cleaning trigger	Risk: Premature/over-frequent cleaning Countermeasure: Set trigger with hysteresis + engineering review before cleaning	22-Jun-2026	STA-02

PM Schedule Established

PM-1: Daily HE-3301 dP & duty trend review (Process, Daily)

PM-2: Bundle inspection & cleaning at dP trigger / next TA (Static, Condition / TA)

Impact vs Control Prioritization

CONTROL / IMPACT	Low	Medium	High
Low			
Medium			X1 X2 X3
High			P1 P2 P3

Downtime & Closure Summary

DOWNTIME 12.0 h	PRODUCTION LOSS 216.0 ton	ESTIMATED LOSS \$183.6k	PRE-RISK III (400)
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KEY CORRECTIVE / PRO-ACTIVE ACTIONS

- Add dP-based cleaning trigger & alarm for HE-3301
- Tighten feed heavy-ends control & filtration
- Add fouling factor & duty KPI to reliability dashboard